

Project 1 — Flow & Tech Stack

Equity & Credit Research Copilot with Numeric Verification

The governing rule of the whole design: narrative comes from retrieved text, numbers come from a structured store, and the LLM never does arithmetic.

FLOW A

Ingestion — offline, scheduled

- 1. Acquisition.** Poll EDGAR's daily index files for new filings by form type (10-K, 10-Q, 8-K) against a CIK watchlist. Separately pull the quarterly XBRL Financial Statement Data Sets in bulk. Idempotent by accession number, so re-runs never duplicate. *Note on transcripts:* most earnings-call transcript sources are licensed, so build on filings first and treat transcripts as an optional extension — do not let a licensing question block the pipeline.
- 2. Parsing and section segmentation.** Filings arrive as messy inline-XBRL HTML. Strip to clean text, then segment by item boundaries — Item 1A Risk Factors, Item 7 MD&A, Item 7A market risk, Item 8 financial statements and notes. This is the step most RAG projects skip and the one that most improves retrieval quality, because "what are their risks" should never retrieve from MD&A.
- 3. Chunking.** Parent-child: embed small chunks (roughly 400-600 tokens) for retrieval precision, but return the enclosing section for generation context. Preserve tables as structured blocks rather than flattening them into prose.
- 4. Metadata tagging.** Every chunk carries CIK, ticker, form type, fiscal period, filing date, section ID, and accession number. This is what makes filtered retrieval possible, and filtered retrieval is what makes multi-company comparison work.
- 5. Two destinations — the architectural spine.** Chunks and embeddings go to the vector store. XBRL facts — every tagged financial figure with element name, period, unit, and source accession — go to a relational table. Every number in the final output will be a lookup against that table, with a fact ID you can trace.
- 6. Versioning.** Embedding model, chunker version, and parser version are stamped on every row, so a re-embed is a controlled migration rather than a rebuild.

FLOW B

Query — online

- 1. Intake and scope resolution.** Resolve entities and periods deterministically before any model call — ticker to CIK against a lookup table, "last three years" to concrete fiscal periods. Letting an LLM guess a CIK is a silent failure source.
- 2. Plan.** The planner decomposes the question into sub-questions and tags each as *narrative* or *numeric*. It emits a typed plan object, not free text. That plan is your audit trail and your debugging surface.
- 3. Route — the triage layer.** Simple numeric lookups ("what was their FY24 revenue") go straight to the structured store with a small model and no retrieval at all. Narrative and multi-company synthesis take the full path. The cost difference between the two paths is your cost-per-query curve.
- 4. Retrieve.** Hybrid search — BM25 for exact terms (ticker symbols, accounting line items, defined terms) plus dense vectors for semantics, with a metadata pre-filter on CIK, period, and section. Rerank with a cross-encoder over the top ~50, keep the top ~8, expand each to its parent section.
- 5. Compute.** The calculator agent does not output numbers — it outputs a *computation spec*: operands referencing specific XBRL fact IDs plus an operation. A sandboxed executor resolves the facts and computes. If a required fact is missing it fails loudly rather than letting the model improvise a plausible figure.
- 6. Draft.** The writer composes prose with inline citations to accession number plus section, and references computed values by ID.
- 7. Verify — the cheap edge.** A separate pass walks the draft, re-derives every numeric claim against the fact table, and confirms every assertion carries a citation. Anything unresolvable is stripped or flagged, never silently published. The verifier emits a report: figures checked, figures traced, figures rejected. **That report is your headline metric.**
- 8. Persist the trace.** Every stage span, token count, and cost lands in the observability store.

STACK

Technology choices, with the reasoning

Orchestration — LangGraph. Stateful graph, checkpointing, deterministic control flow, native human-in-the-loop interrupts: exactly what an auditable pipeline needs. Avoid AutoGen, which moved to maintenance mode in favour of the Microsoft Agent Framework — building on a maintenance-mode framework is a question you do not want in your review.

Models — tiered. A small fast model for routing, entity resolution, and classification; a frontier model for planning, synthesis, and verification. Access through a single provider-abstraction layer so you can swap and A/B without touching graph code. Monitoring should show which tier handled what share of traffic.

Vector store — pgvector on Postgres. The 2026 practitioner consensus makes pgvector the default for new RAG projects; it handles tens of millions of vectors and avoids running a second database. For this project it is doubly right, because XBRL facts, the filing registry, and chunks then live in **one database** — joining a citation to a computed figure becomes a SQL join instead of a distributed lookup. Qdrant is the alternative if filtered-query latency bites: pgvector post-filters, which can under-retrieve when metadata filters are aggressive, whereas Qdrant filters in-graph. Put that trade-off in your design doc; knowing *why* you chose is worth more than the choice.

Lexical search — Postgres full-text search. Keeps the single-database story intact. OpenSearch only if you genuinely need it.

Reranker — cross-encoder. bge-reranker-v2 self-hosted, or a hosted rerank API. The single highest-leverage retrieval improvement per hour spent.

Sandbox. Restricted Python execution: no network, no filesystem, hard timeout. Keep fact-resolution logic outside the sandbox and pass in resolved values.

Application layer — FastAPI + Pydantic v2. Typed contracts between agents; typed plan objects and computation specs are what make the pipeline debuggable. Async workers via Arq or Celery on Redis for long queries.

Ingestion orchestration — Prefect or Dagster. Preferable to Airflow at this size: less ceremony, better local development.

Frontend — Streamlit or Next.js. Streamlit if you want the time back for the verifier and eval harness; Next.js if demo polish matters to your guide. Be honest with yourself about which.

Observability — OpenTelemetry GenAI conventions. Instrument to the open spec, with Langfuse or Arize Phoenix self-hosted as the trace UI. Track per-stage latency, tokens, cost, retrieval hit-rate, citation coverage, and verifier rejection rate. Using the open spec rather than a vendor SDK is a small choice that signals production experience.

Evaluation and CI/CD. A golden set of 50–100 questions with known answers, run through Ragas-style retrieval metrics plus your own numeric-accuracy check, with GitHub Actions gating merges on it. This covers the Test Cases rubric row properly, which most of the cohort will not.

Deployment. Docker, Terraform for infrastructure, container hosting on Cloud Run or ECS. Kubernetes is unnecessary overhead at this scale and will cost you a week that belongs elsewhere. Managed Postgres with the pgvector extension.

Security. Secrets in a cloud secret manager, OIDC auth, per-tenant row-level security in Postgres, and — the demonstrable one — a prompt-injection test suite. Plant instructions inside a filing, show the naive pipeline following them, then show your defence: structured tool contracts, no raw document text in system prompts, output validation.

Build order

Weeks	Focus
1-3	Ingestion and retrieval. Everything downstream depends on chunk quality, so this comes first.
4-5	Numeric layer and structured store. This is the differentiator — give it room.
6-7	Agent graph: planner, router, retrieval agent, calculator, writer.
8-9	Verification pass and evaluation harness.
10-12	Deployment, monitoring, security, and the prompt-injection demonstration.

If you run short on time, cut the transcript source and the polished frontend. **Do not cut the verifier or the eval harness** — they are the two things that make this distinguishable from every other Agentic-RAG assistant in the room.

Headline metric to lead the presentation with: hallucinated-figure rate before and after the verification pass, alongside cost per query split by triage path. Data sources are all free — SEC EDGAR full-text and daily index, XBRL Financial Statement Data Sets, yfinance for prices.